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Experimental Evidence of CO₂ Photoreduction Activity of SnO₂ Nanoparticles

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Supporting information for this article is given via a link at the end of the document.

Abstract: Tin dioxide (SnO₂) has intrinsic characteristics that do not favor its photocatalytic activity. However, we evidenced that surface modification can positively influence its performance for CO₂ photoreduction in the gas phase. The hydroxylation of the SnO₂ surface played a role in the CO₂ affinity decreasing its reduction potential. The results showed that a certain selectivity for methane (CH₄), carbon monoxide (CO), and ethylene (C₂H₄) is related to different SnO₂ hydrothermal annealing. The best performance was seen for SnO₂ annealed at 150°C, with a production of 20.4 μmol g⁻¹ for CH₄ and 16.45 μmol g⁻¹ for CO, while for SnO₂ at 200°C the system produced more C₂H₄, probably due to a decrease of surface -OH groups.

Several semiconductors, highlighted by TiO₂, have photocatalytic activity to promote redox reactions, such as the oxidation of organic substances or reduction of molecules such as CO₂ or N₂. These reactions occur through photoelectrochemical processes, driven by the charge separation during illumination and specific electronic structure of each material.^[1] SnO₂ has an electronic structure similar to TiO₂, but its electronic mobility is almost 100 times higher, leading to high efficiency in the photogenerated electrons transference.^[2] On the other hand, the position of its conduction band minimum (CBM) is located above the redox potential level of H⁺/H₂ (-0.41 V vs. NHE, at pH 7),^[3] making SnO₂ limited for application in reduction reactions. This feature also limits its application in photocatalytic oxidation since the reaction is controlled by the coupled reduction reaction.

Electrochemical experiments indicate that SnO₂ nanoparticles are efficient electrocatalysts of CO₂ reduction due to the interaction of carbonate species with surface Sn hydroxides.^[4] In this case, the energy for reduction is externally supplied, and analysis of reduction potential for SnO₂ indicates that the oxide is not reduced to Sn in the necessary potential range for CO₂ reduction, keeping stable. On the other hand, Sn hydroxides may have lower reduction potentials, favoring the redox process.^[5] This surface interaction is expected to occur in photocatalytic systems since no bias is necessary to promote it, but the reaction would be limited by the band's energy.^[6] Therefore, to overcome the limitation, the hydroxylation of SnO₂ can, in principle, acts as a charge mediator, reducing the reduction potential to favorable ranges for CO₂ reduction.^[4,7] Sn(OH)₄, the most probable

hydroxide in sol-gel synthesized SnO₂ surfaces, is expected to be reduced to Sn(OH)₂ at lower potentials than H⁺/H₂ and CO₂/CO. Other hydroxides can be formed depending on the synthesis condition, which can be modified by crystallization routes – such as hydrothermal annealing.^[8]

Therefore, herein we demonstrate that SnO₂ is a potential photocatalyst for CO₂ reduction, where the surface hydroxylation plays a role in the reduction potentials (as charge mediator) and the CO₂ affinity. We modified the hydroxylation degree of co-precipitated SnO₂ (sol-gel synthesis) by different hydrothermal annealing temperatures, which evidenced that the increase of strongly bonded -OH groups promotes the reduction efficiency. On the other hand, the higher annealing temperature is deleterious, probably due to the higher crystallization of these hydroxides. However, this crystallization modifies the product selectivity, slightly increasing the reduction to C₂ products. These results reveal the potential of SnO₂ that merits deeper investigation for these applications.

SnO₂ nanoparticles were synthesized at room temperature and annealed via hydrothermal conditions at different temperatures. In detail, the SnO₂ was synthesized as follows (Fig. S1): 0.564 g of SnCl₂·2H₂O (Sigma-Aldrich) was dissolved in 100 mL of ethanol anhydrous (99.5% purity, Dinâmica) and then under vigorous stirring, 22.5 mL of distilled water was added dropwise at room temperature for 12 h.^[9,10] After that, the suspension was cleaned by dialysis in 5 L of distilled water, often renewed, until removing all chloride ions, which was confirmed by tests with AgNO₃ solution (0.1 mol L⁻¹).^[11] Then, the solution inside the dialysis bag was dried at 70°C for 12 h, ground with an agate mortar, and named "SnO₂-cop". After that, the as-prepared sample was treated hydrothermally up to 200°C for 4 h.

The characterization procedure using different techniques, such as X-ray diffractometry, scanning electron and high-resolution transmission microscopy, Near-Infrared Spectroscopy, and diffuse reflectance spectroscopy are shown in detail in the **Supporting Information**. The catalysts have been characterized in CO₂ photoreduction experiments, carried out in a gas-solid reaction using CO₂ + H₂O steam for 6 h under UV light. The catalysts were deposited over a glass substrate,

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making a thin film with 10 mg and 3.24 cm² of the exposed area. The reactants have been supplied as a moist CO₂ by bubbling in water. A scheme of the experimental setup is shown in Suppl. Fig. S2.

The morphology and size of the SnO₂-cop particles were examined by scanning electron microscopy (SEM) and transmission electron microscopy (TEM) (Fig. 1a-b). SnO₂-cop sample is precipitated in very massive agglomerates, consisting of large and faceted xerogel blocks, from 30 and 60 μm (Fig. 1a). The xerogel structure is supported by the high surface area (188 m² g⁻¹) and composed of small crystalline particles (Figure 1b) of 4 nm, with adjacent lattice spacing (0.34 nm) referring to the planes (110).^[12,13] Line defects (white arrows), typical of oriented attachment growth, are seen evidencing the growth mechanism. The X-ray diffraction patterns (Fig. 1c) indicate that the hydrothermal annealing did not affect the phase, all indexed as tetragonal cassiterite phase (JCPDS No. 41-1445). The wide peaks and low intensity are characteristics of the amorphous nature of the samples.^[14,15] The peak sharpening for the sample at 200°C is associated with the oriented growth of SnO₂ the particles, as expected.^[16] The *bandgap* value (Fig. S3) found for the commercial sample (3.7 eV) agrees with the literature, while to the other samples, the values were lower.^[17,18] The bandgap values for the SnO₂-cop and SnO₂-200 are 3.2 and 3.5 eV, respectively, a slight blueshift in the *bandgap* energy.

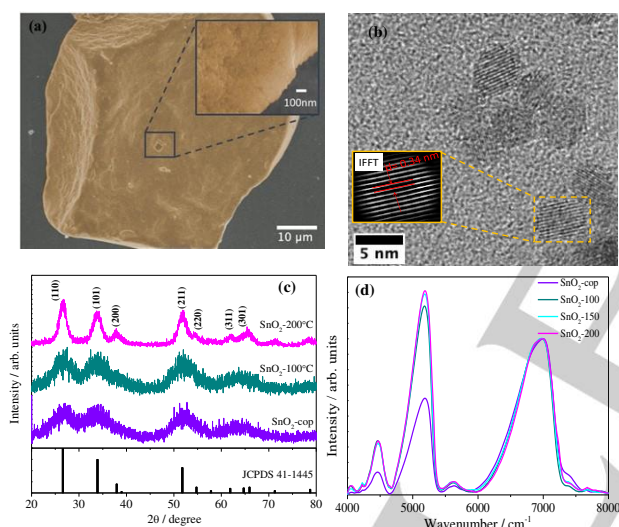


Figure 1. (a) SEM images; (b) HRTEM images; (c) XRD patterns of the synthesized SnO₂ samples, and (d) Near-infrared spectrum (NIR) of the SnO₂-cop and SnO₂ submitted to hydrothermal annealing at 100, 150, and 200°C.

The SnO₂ utilization for photocatalytic oxidation is barely described but generally is reported as inactive. However, surface groups directly influence photoactivity, as seen in our experiment (Fig. S7): the surface modification of SnO₂ through hydrothermal treatment was effective since expressive results were obtained, mainly with the SnO₂-200 sample. NIR analysis (Fig. 1d) contributes to elucidating the mechanism involved during the oxidation process, being directly related to the water molecules adsorbed on the surface of the SnO₂ samples. As seen in Figure 1d there are three characteristic peaks of water, i.e., ~7,000 cm⁻¹ corresponds to surface hydroxyl groups (1st overtone), ~5,000 cm⁻¹ attributed to crystalline water (hydroxide, 1st overtone) and 4,500 cm⁻¹ attributed to xerogel structure.^[19] Normalizing all the peaks by the surface hydroxyl groups, one can see differences in the distribution of crystalline water, which tends to

increase (in proportion) according to the annealing temperature. It is attributed to the elimination of weak-bonded water, which is expected. A deeper analysis of the bonded-water distribution is seen by deconvoluting the crystalline water in four contributions, as seen in Figure S4. Table 1 summarizes the proportion of each contribution, which are attributed to different water vibration modes S₁, S₂ and S_n. The temperature increases the contribution of strongly bonded water (S₁) by reducing polymerized water (S_n). It indicates that the water-water interaction diminishes with temperature, but one can see that the distribution of water bonding modes is almost unaltered from SnO₂-cop to SnO₂-100, indicating that, in this range, the only modification is the elimination of weak-bonded water.

Table 1. Relative area (%) of the deconvolution band from NIR spectra of the synthesized samples.

Samples	Vibration mode of the water molecules				
	S ₁	S ₂	S _n		
	Peak ₁	Peak ₂	Peak ₁ + Peak ₂	Peak ₃	Peak ₄
SnO ₂ -cop	45.62	8.21	53.83	32.64	13.53
SnO ₂ -100	47.33	6.83	54.16	32.60	13.24
SnO ₂ -150	48.22	8.85	57.07	31.49	11.44
SnO ₂ -200	46.10	14.01	60.11	32.55	7.34

S₁ – one active H-bond; S₂ – two active H-bond; S_n – H₂O with two active H-bond and one (or two) passive H-bond (polymeric chain of H₂O molecules).

A comparison of the total number of water groups at surface is done by normalizing the FTIR spectrum by the O-Sn-O peak area (A_{O-Sn-O}, 581 cm⁻¹),^[20] which is considered to be constant for all samples (since the composition is the same). Therefore, the intensity of H₂O and OH peaks (A_{H₂O} and A_{OH}) will vary according to the relative amounts of water molecules, which are clearly seen by the quotient A_{H₂O}/A_{O-Sn-O} and A_{OH}/A_{Sn} (Table 2). Thus, the contribution of OH groups (weakly bonded) for SnO₂-cop is twice the observed for SnO₂-200, indicating the water elimination. These data also reveal that, despite the distribution of water group modes is similar for SnO₂-cop and SnO₂-100 (as seen in NIR experiments), the content of weakly bonded and crystalline water in SnO₂-cop is much higher. Besides, SnO₂-150 seem to be intermediary, showing almost the same weakly bonded water content of SnO₂-100 but with more crystalline water, which is probably from the surface crystallization of hydroxyl groups (Fig. 1d). Finally, SnO₂-200 has lower content of both weakly-bonded and crystalline water since the temperature is probably leading to the crystallization of surface oxide (anhydrous).

This variation correlates with the CO₂ reduction profiles, suggesting that the crystalline water plays a role in the reaction. SnO₂-100, which has a reduction of crystalline water content, led to lower

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photoactivity compared to the others. SnO₂-150, having the highest content of crystalline water and less polymeric water, produced to the best photoactivity, which is dramatically reduced in SnO₂-200 due to the lower content of water groups. Therefore, we suggest that the Sn hydroxyl groups are the actual catalytic sites, which are influenced by the bonding mode and by the total content. This hypothesis indicates that the photocatalytic trend SnO₂-150 > SnO₂-cop > SnO₂-100 is due to two factors, the crystallization water at surface and the total water content.

Table 2. Absolute Areas (A) of the FTIR spectrum bands of the synthesized samples.

Samples	A _{O-Sn-O} (581 cm ⁻¹)	A _{H₂O} (1630-1640 cm ⁻¹)	A _{OH} (2500-3500 cm ⁻¹)	A _{H₂O/A_{O-Sn-O}}	A _{OH/A_{O-Sn-O}}
SnO ₂ -cop	42.00	4.44	44.90	0.11	1.07
SnO ₂ -100	65.00	3.65	28.92	0.06	0.44
SnO ₂ -150	91.77	5.30	46.98	0.06	0.51
SnO ₂ -200	100.76	3.98	42.63	0.04	0.42

These molecules are possibly regarded to the surface hydroxylation of SnO₂, coordinated as Sn(OH)₄, as a remnant of polycondensation step during coprecipitation.^[21] These OH groups undergo oxidation by the holes (*h*⁺) formed in the catalyst valence band when under UV, leading to the formation of hydroxyl radicals ($\bullet$ OH) with high oxidizing power ($-\text{SnOH} + h^+ \leftrightarrow -\text{Sn}^{2+} + \bullet\text{OH}$).^[22] Adsorbed OH- groups on the catalyst surface this potential to 1.6V vs NHE, favoring the radical formation.^[3] Therefore, the attack of these radicals leads to oxidation reactions, indicating that the indirect photodegradation mechanism plays an important role in the photoactivity of the synthesized SnO₂ samples.^[21,23]

Fig. 2a indicates the potential of SnO₂ samples to photoreduce CO₂ to C1 species. Although the intrinsic characteristics of SnO₂ are not favorable to its photoactivity, these results reveal that the conditions used in the reactions were probably essential for selective response observed to certain products. The samples were selective for three different products, CH₄, CO, and C₂H₄, respectively (Fig. 2a), with higher CH₄ selectivity for all samples except for 200°C-annealed (Fig. 2c). Fig. 2b shows the number of electrons transferred for each product^[24] according to Equation 1 (see in *Supporting Information*), knowing that the transference of 8*e*⁻ are necessary for the production of CH₄ from CO₂ reduction, 2*e*⁻ for CO and 12*e*⁻ for C₂H₄. The gas-phase reactor implies that the CO₂ concentration is extremely high, considering that humidified, high purity CO₂ is purged within the system for 20 min. Thus, taking into account the acidic nature of CO₂ molecules, its adsorption will reduce the local pH, consequently lowering the potential for CO₂ reduction.^[25] To determine the acid-base nature of the SnO₂ surface, the isoelectric point of sample suspensions at different pH values (Figure S6) was determined, which predicts the adsorption behavior of CO₂ molecules during the photoreduction process. As expected, although the values found were close for all samples and in accordance with the literature, a small deviation to alkaline behavior is seen to SnO₂-100, due to the reduction of crystalline surface water (as evidenced by FTIR measurements), which favors that this reaction can take place in lower local pH since below IEP its surface is positively charged.^{[7,10].}

Moreover, SnO₂-150 has the more acidic surface, corresponding to its highest CO₂ conversion. These values show that the samples have an acidic surface, thus favoring CO₂ adsorption, one of the critical stages of the reaction. The production of CH₄ and CO, in turn, was close (20.4 μmol g⁻¹, and 16.45 μmol g⁻¹, respectively, for SnO₂-150). This result is noteworthy when compared to the few reports in the literature about SnO₂ photoreduction activity, even also to representative TiO₂ systems (Table 3).

However, in addition to CH₄ and CO, we also observed the production of small amounts of C₂H₄ (Fig. 2c). An increase in C₂H₄ selectivity is seen for 200°C-annealed sample, considering the total catalytic activity. We hypothesize that the higher annealing temperature has led to higher growth and reduction of surface -OH groups, which is consistent with our results.^[26] It is worth to mention that these results were obtained with pristine SnO₂ and using a low-power lamp (18 W).

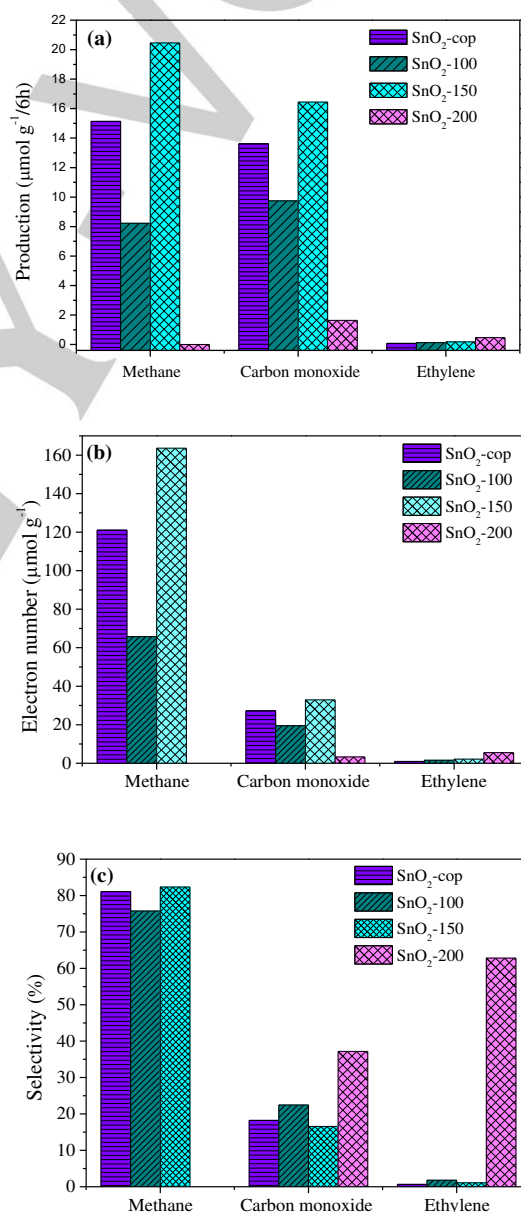


Figure 2. (a) Amount of C-species produced from CO₂ photoreduction driven by synthesized SnO₂ samples activated under UV light in a gas-solid reactor for 6 hours; (b) the corresponding photoelectron transfer number involved in the CO₂ photoreduction and (c) selectivity for CH₄, CO, and C₂H₄ for the different synthesized samples.

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Table 3. Comparison of CO₂ photoreduction production promoted by Sn-based catalysts and TiO₂ a standard catalyst used in photochemical reactions.

Catalyst	Yield $\mu\text{mol g}^{-1} \text{h}^{-1}$				Conditions	Ref
	CH ₄	CO	CH ₃ OH	C ₂ H ₄		
SnO ₂	3x10 ⁻³	—	6.5x10 ⁻⁵	—	450 W	[27]
SnO ₂	1.37	1.14	—	—	300 W	[28]
SnS ₂ /SnO ₂	—	48.01	—	—	300 W (filter $\lambda > 400$ nm)	[29]
SnO ₂ -C ₃ N ₄	0.62	0.72	—	—	300 W Xe	[30]
P25	0.16	0.53	—	—	300 W	[28]
TiO ₂	0.13	0.61	—	—	5 W	[31]
SnO ₂ /Fe ₂ O ₃	5.0	11.25	—	—	300 W (filter $\lambda > 420$ nm)	[32]
SnO₂-150	3.41	2.74	—	0.03	18 W	This work

* These data show the best results for the cited reports.

These trends are consistent with the hypothesis that the surface Sn(OH)₄ groups are playing a role as charge mediators.^[4,26] As observed, the hydrothermal annealing is expected to increase the surface hydroxylation up to 150°C. Despite some divergence between SnO₂-cop and 100°C, which is explained by possible large variations of free water in these conditions, there is an increase in photoreduction activity at 150°C coinciding with a higher amount of hydroxyl groups. It is also noteworthy that the product selectivity is almost identical for these three conditions, as seen in Fig. 2c, which indicates that the reduction mechanism is probably the same. However, the higher crystallization for the sample annealed at 200°C has probably affected the distribution of hydroxyl groups, leading to more ordered hydroxides. These influenced the selectivity leading to C-H bonds (methane and ethylene) rather than C-O. These features are still a matter of investigation, but they evidence the role of surface hydroxylation in SnO₂ photoreduction activity.

In conclusion, we demonstrated that SnO₂ could be photocatalytically active for CO₂ reduction in the solid-gas system (CO₂-water gaseous mixture) under some synthesis conditions, easily modified by hydrothermal annealing. Our results indicate that the structural surface water (surface hydroxylation) plays a role in its photoactivity, and product selectivity can be tuned by modifying the nature of these surface hydroxides by crystallization. Moreover, the catalyst efficiency was higher than previously reported papers, including compared to TiO₂. These results demonstrate that SnO₂ should be more studied for CO₂ reduction due to its advantages, as the high chemical stability, easy synthesis of small and uniform nanoparticles, and good electronic properties.

Acknowledgements

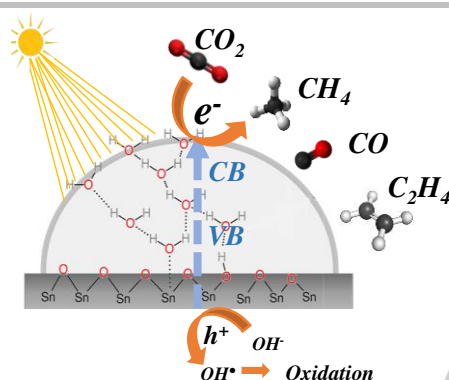
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SnO₂ nanoparticles were synthesized and applied in the CO₂ photoreduction process. The synthesis methodology produced strongly bonded -OH groups, which influenced the process of reducing the CO₂ potential, as well as its affinity. The hydroxyls act as charge mediators, reducing the potential for CO₂ reduction. The semiconductors showed selectivity to C1 hydrocarbons and the sample subjected to higher annealing showed selectivity for C2 products.